Examiner only

5. Satellites can be positioned in orbit to communicate with base stations on the ground. Messages can be relayed around the world in this way. The diagram shows base stations **A**, **B**, **C** and **D** and four satellites.

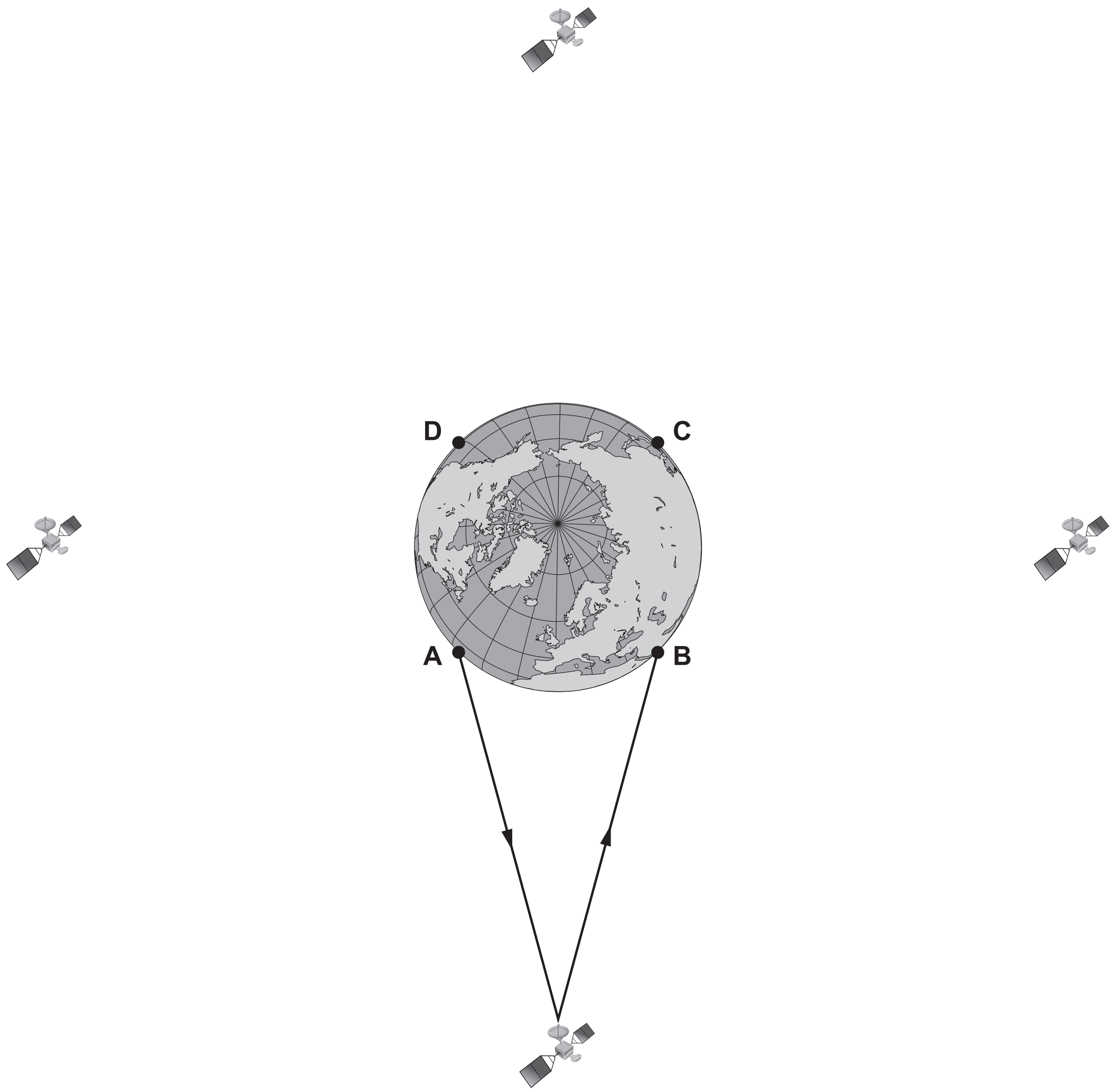

Diagram is not drawn to scale

(a) The diagram shows the path of a signal sent from base station **A** to base station **B**. **Continue the path of the signal** so that it can be received at base station **C**. [1]

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(b) The distance from a base station to a satellite is 36 000 km. Calculate the total distance travelled by the signal in going from **A** to **C**.
Give your answer in **metres**. [3]

Total distance = ... m

(c) The time taken for the signal to travel from **A** to **C** is 0.48 s. Use an equation from page 2 to calculate the speed of the signal. [2]

Speed = ... m/s

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